Report

Upgrade Audit

by Asymptotic

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Subject: Independent review of the changes deployed on top of the audited baseline (on-chain package version 17).

What was reviewed

- **Package:** `bluefin_spot` — on-chain package **version 17**, owner `Immutable`, verified against the Sui mainnet fullnode.
- **Baseline:** the commit covered by Asymptotic's manual audit report.
- **Delta:** every source change shipped between that baseline and the current mainnet deployment.

Nature of the upgrade

This release is a **dependency-migration release**, undertaken in response to the May-2025 Cetus exploit. The deployment removes the affected third-party arithmetic dependency entirely and re-routes all signed-integer math through a Bluefin-maintained library, behind a defensively written conversion layer that includes an explicit kill-switch against the upstream bug. On-chain object layouts are preserved, so the upgrade is binary-compatible. The core trading, accounting, and oracle logic is the established CLMM math, unchanged.

The accompanying changes are small and self-contained: a composable two-step pool-creation path, an admin-gated pool icon-URL setter (with a no-op guard and a new event), and event/visibility cleanups. The router's exact-out path now follows standard Uniswap-V3 partial-fill semantics; the caller's input-side slippage bound is retained.

Conclusion

We identified **no findings at any severity**. The migration is sound, on-chain object compatibility is preserved, and the supporting changes

introduce no new permissionless attack surface.

This report reflects our review of the code changes at the commits and on-chain versions noted above. A review reduces the risk of certain classes of bugs but does not guarantee the absence of all vulnerabilities; only the reviewed changes and properties are covered. This report is provided for informational purposes only, not as financial, legal, or investment advice. No liability is assumed for losses arising from reliance on this report. Users should perform their own due diligence before interacting with any smart contract system.